EE/CprE/SE 491 WEEKLY REPORT 6

Apr 2, 2025* - *Apr 8, 2025

Group number: *16*

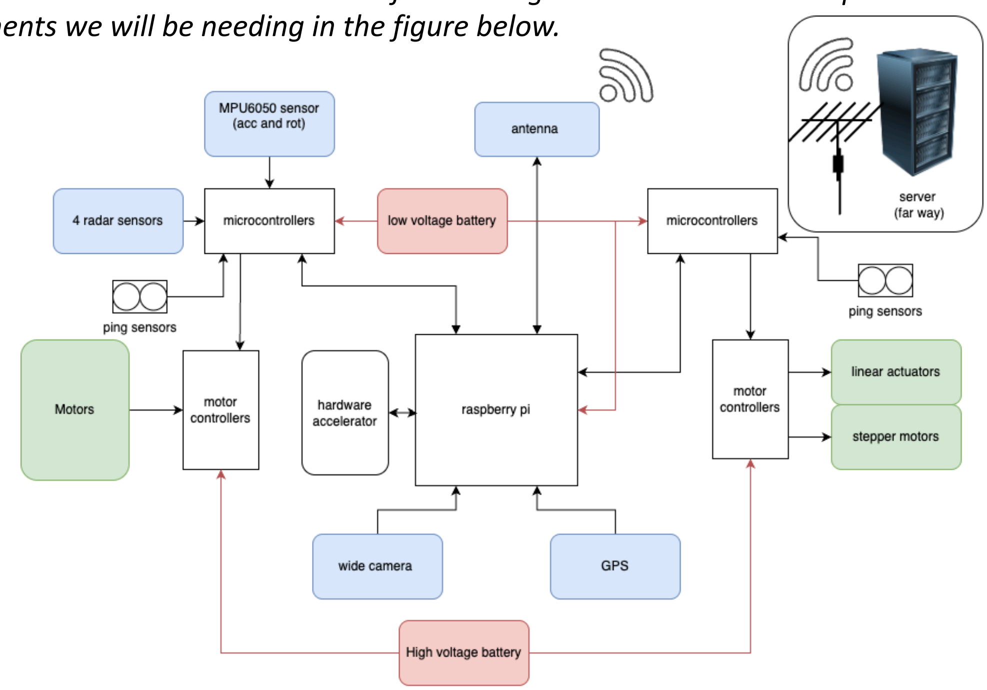
Project title: *Outdoor Roomba*

Client &/Advisor: *Prof. Diane Rover*

Team Members/Role:
Brodie M Bates
Daniel Vergara Pinilla
Mitchell D Owen
Om Shukla
Sudipta Halder

- **Weekly Summary**

This week due to major price changes we have begun working towards finalizing our design(for this semester) for our robot and continued working on Bills of Materials. We have looked into microcontrollers and different protocols (SPI, CAD, I2C, etc) that best fits our robot's requirements. Continued to adjust and update our requirements. Initially our requirements for GPS precision was 2 feet or 0.6096 meter of the position but most of the GPS modules had a precision of 3 meters. The ones that can achieve 0.6 precision were out of our budget. So we have increased that radius to better fit our budget. So Far here is our top level electronics comments we will be needing in the figure below.



- **<u>Past week accomplishments</u>**

    - Sudipta Halder: discussed function of the server, what inputs and outputs we need from the robot and the server. Discuss about user inputs and interaction. Microcontrollers and its interaction with the main computer(raspberry pi), worked on completing the BoM, and figured out a new way to make robots in Webots via *Protos* (a file similar to json, a description language for Webots to make costume “Nodes”).

    - Brodie Bates: Test SPI on microcontrollers, CAD chassis of robot, research motors and sensors for the bill of materials.

    - Daniel Pinilla: research on multi-threading

    - Om Shukla: Still researching geofencing and how to implement it into our Java backend and what kind of software systems the robot will need onboard.

    - Mitchell Owen: Found a motor, a motor controller, and benchmarked against a competitor.

- **<u>Pending issues</u>**

    - Mitchell Owen: The biggest issue currently is budgeting. Parts are getting more expensive.

- **Individual contributions**

NAME	Individual Contributions	Hours this week	HOURS cumulative
Brodie Bates	Research sensors and motors to add to bill of materials, Test SPI protocols for sensor controller, CAD chassis of robot.	6	35
Om Shukla	Researched geofencing and robot onboard software systems.	4	27
Sudipta Halder	Started diagram for the electronic components and how the sensor and motor will interacted with each via SPI,Uart, etc.	7	39
Daniel Mauricio Vergara	AI object detection and movement direction	7	32
Mitchell Owen	Researching Webots	5	30

- **Plans for the upcoming week**

  - Sudipta Halder: work on the new Cad model with the sensor and component to figure out the weight, we can determine which motors and battery to use.
  - Brodie Bates: CAD more of the robot, research final servos and battery for the robot, 3D print robot parts.
  - Om Shukla: Learn more about object marking in a geofence and the onboard systems the robot needs for object detection and avoidance. Learn more about geofencing and how it works.
  - Mitchell Owen: Order parts and start working on the chassis if we get everything.
  - Daniel Mauricio Vergara: 3D modeling and working on websocket code for the robot

- **Summary of weekly advisor meeting**

This week the advisor meeting was urging us to complete the bill of materials so we can get parts ordered before the tariffs. We also need to have more communication between the parts of our team server for our server side, physical hardware side, and microcontroller side.